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Semi - Major test - 04

Physics

1) (1) - 1

2) (2) Outer surface of the cornea

3) (4) Virtual, erect, enlarged

4) 1) Both A and R are true and R is the correct explanation of the A.

5) (3) Assertion is true but Reason is false.

6) Two common vision defects are :-

i) Myopia / Near Sightedness.

ii) Hypermetropia / Far-sightedness.

These defects can be fixed by using of suitable and proper spherical lenses.

7) a) When we see an object from nearer to denser medium, then it appears as closer to us than its real/actual depth, it's because of refraction of light and bending of light rays.

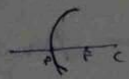
b) Given,

$$m = -2, \quad v = 20 \text{ cm}$$

$$m = \frac{+v}{u}, \quad \Rightarrow -2 = \frac{+20}{u} \Rightarrow (-)u = -20 \Rightarrow u = 10 \text{ cm}$$

~~$$m = -2 = \frac{v}{20} \Rightarrow v = -40 \text{ cm}$$~~

$$\boxed{u = 10 \text{ cm}}$$



8) Given,

$$u = -10 \text{ cm}, f = 15 \text{ cm},$$

using,

$$m = \frac{f}{f - v} = \frac{f - u}{f}$$

$$\frac{15}{15 - v} = \frac{15 - (-10)}{15}$$

$$\Rightarrow \frac{15}{15 - v} = \frac{25}{3}$$

$$\Rightarrow 45 = 75 - 5v$$

$$\Rightarrow 75 - 5v = 45$$

$$\Rightarrow 75 - 45 = 5v$$

$$\Rightarrow 30 = 5v$$

$$\Rightarrow v = \frac{30}{5} = 6$$

$$\Rightarrow \boxed{v = +6 \text{ cm}}$$

$\therefore$ The object is formed 6 cm behind the mirror and its virtual and erect.

9) Given,

f

$$u = -10 \text{ cm}, v = ? \text{ cm}, m = -3$$

using,

$$m = + \frac{v}{u}$$

$$-3 = \frac{v}{-10}$$

$$\boxed{v = 30 \text{ cm}}$$

$\therefore$ The

the object

Now, to find focal length,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}, \quad \frac{1}{30} - \frac{1}{-10} = \frac{1}{f}$$

$$\frac{1}{30} - \frac{1}{\infty} = \frac{1}{f}$$

$$\frac{1-3}{30} = \frac{1}{f}$$

$$\frac{21}{30} = \frac{1}{f}$$

$$\Rightarrow \boxed{f = 15 \text{ cm}}$$

∴ The image distance is 30 cm.
and the focal length is 15 cm.

10) Given, far point of a myopic person = 80 cm

So, the nature of the lens used to fix a myopic person is concave lens.

$$f = -80 \text{ cm} = -0.8 \text{ m}$$

$$P = \frac{1}{f}$$

$$P = \frac{1}{-0.8 \text{ m}} \Rightarrow \boxed{P = -1.25 \text{ Diopter}}$$

So, the nature is concave lens and power is -1.25 Diopter.

11) a) Convex lens.

b)

A/Q,

$$m = \frac{v}{u}$$

$$-500 = \frac{10}{u} \Rightarrow u = \frac{10}{-500} = -\frac{1}{50}$$

$$\Rightarrow u = -\frac{1}{50}$$

The sign will be negative as because
by solving it came negative, also because
the image formed is real and inverted.

$$m = -500 = \frac{10}{\frac{1}{50}}$$

$$-f = 500 = 10 \times 50$$

c) Given, $f = 20$, $u = 21$ cm,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{20} = \frac{1}{v} - \frac{1}{21}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{1}{21}$$

$$\Rightarrow \frac{1}{v} = \frac{21 - 20}{420}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{420}$$

$$\Rightarrow \boxed{v = 420 \text{ cm}}$$

12) a) i) The incident ray, the refracted ray and the normal at the point of incidence all lies in the same plane.

ii) The sin of angle of incidence to the sin of angle of refraction gives a constant for two medium 1 and 2.

$$\frac{\sin i}{\sin r} = \text{constant}$$

b) Given:

~~refractive index~~ Expression to relate absolute refractive index:-

$$n = \frac{c}{v}$$

absolute refractive index = $\frac{\text{Speed of light in vacuum}}{\text{Speed of light in medium}}$

b) Refractive index of water = $\frac{4}{3}$
 " " " glass = $\frac{3}{2}$

Speed of light in glass = $2 \times 10^8 \text{ m/s}$

Speed of light in

(i) air = $n = \frac{c}{v}$

$$\frac{3}{2} = \frac{c}{2 \times 10^8}$$

$$3c = 3 \times 2 \times 10^8 \quad \Rightarrow \quad c = 3 \times 10^8 \text{ m/s.}$$

ii) $n = \frac{c}{v}$

$$\frac{4}{3} = \frac{3 \times 10^8}{v} \quad \Rightarrow \quad v = \frac{3 \times 3 \times 10^8}{4}$$

$$\Rightarrow v = 2.25 \times 10^8 \text{ m/s}$$

$$\begin{array}{r} 2.25 \\ 4 \overline{) 9} \\ \underline{8} \\ 10 \\ \underline{8} \\ 20 \\ \underline{20} \\ 0 \end{array}$$